



EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN

WEEK 8

Assignment: Object-Oriented Analysis — Behavioural Modelling

PROGRAMME

BSc Information Communication and Technology

COURSE CODE / YEAR

BICT 3202 · Year 3

LECTURER

Francis Fweta — ICT Lecturer

DELIVERY

2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

Prepared by Francis Fweta · Exploits University · Week 8 of 16



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ASSIGNMENT INSTRUCTIONS

- Answer ALL questions in Sections A, B and C.
- Read the Week 8 module and your lecture notes before attempting each question.
- Draw diagrams neatly (by hand or using a UML/CASE tool) with correct notation.
- Label all actors, use cases, relationships and multiplicities.
- Submit by the date given by your lecturer. Total: 50 marks.

Section A — Short-Answer Questions (20 marks)

Question A1

Define a use case and explain its purpose in requirements capture. [2 marks]

Question A2

Define an actor and differentiate it from a user. [2 marks]

Question A3

Explain the difference between a primary actor and a secondary actor. [2 marks]

Question A4

What is a use-case diagram? What are its main elements? [2 marks]

Question A5

Define an interaction diagram and name the two main types. [2 marks]

Question A6

Explain the purpose of a sequence diagram. [2 marks]

Question A7

What is a collaboration diagram? How does it differ from a sequence diagram? [3 marks]

Question A8

Define an activity diagram and explain when it should be used. [2 marks]

Question A9

Explain the relationship between use cases and scenarios. [3 marks]

Section B — Application Questions (20 marks)

Question B1 (12 marks)

'A university wants an online course registration system. Students search for courses, select courses, submit registration and receive confirmation.'

- (a) Identify the actors. [2 marks]
- (b) Identify the use cases. [2 marks]
- (c) Draw a use-case diagram. [4 marks]
- (d) Write the main scenario for 'Register for Course'. [4 marks]

Question B2 (8 marks)

Consider a student registering for a course with prerequisites.

- (a) Draw a sequence diagram for successful registration. [4 marks]
- (b) Explain what happens if prerequisites are not met. [4 marks]

Section C — Case Study (10 marks)

'A library wants a book borrowing system. Members search for books, reserve books, borrow books and return books. Librarians manage the book collection.'

As an object-oriented analyst:

- (a) Draw a use-case diagram for the library system. [4 marks]
- (b) Draw a sequence diagram for the 'Borrow Book' use case. [3 marks]
- (c) Explain how these diagrams help in understanding system behaviour. [3 marks]

Marking Scheme

Section	Content	Marks
A	Short-answer questions (9 questions)	20
B	Application questions (12 + 8)	20
C	Case study — library system	10
	TOTAL	50

Submission & Academic Integrity

- Submit a typed document (or neatly handwritten, as directed by your lecturer).
- This assignment is individual work. Plagiarism or copying will attract a penalty under university regulations.
- Diagrams must use correct UML notation.
- Cite any sources you consult using the format used in the Week 8 module references.